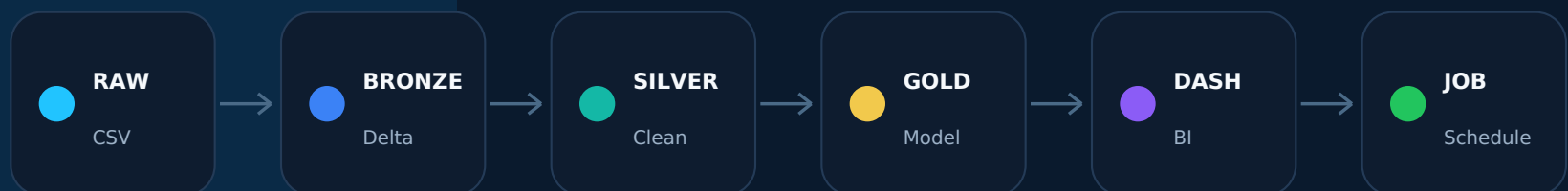


SAP SD Sales Analytics

Databricks Medallion Data Engineering Project

An end-to-end project covering raw ingestion, data quality engineering, business modeling, dashboarding, orchestration, scheduling and monitoring.



Omar Oun

Built as a hands-on data engineering portfolio project

PySpark

Delta

Unity Catalog

AI/BI Dashboard

Lakeflow Jobs

SAP SD

Sample/mock SAP-style data used for learning and portfolio purposes.

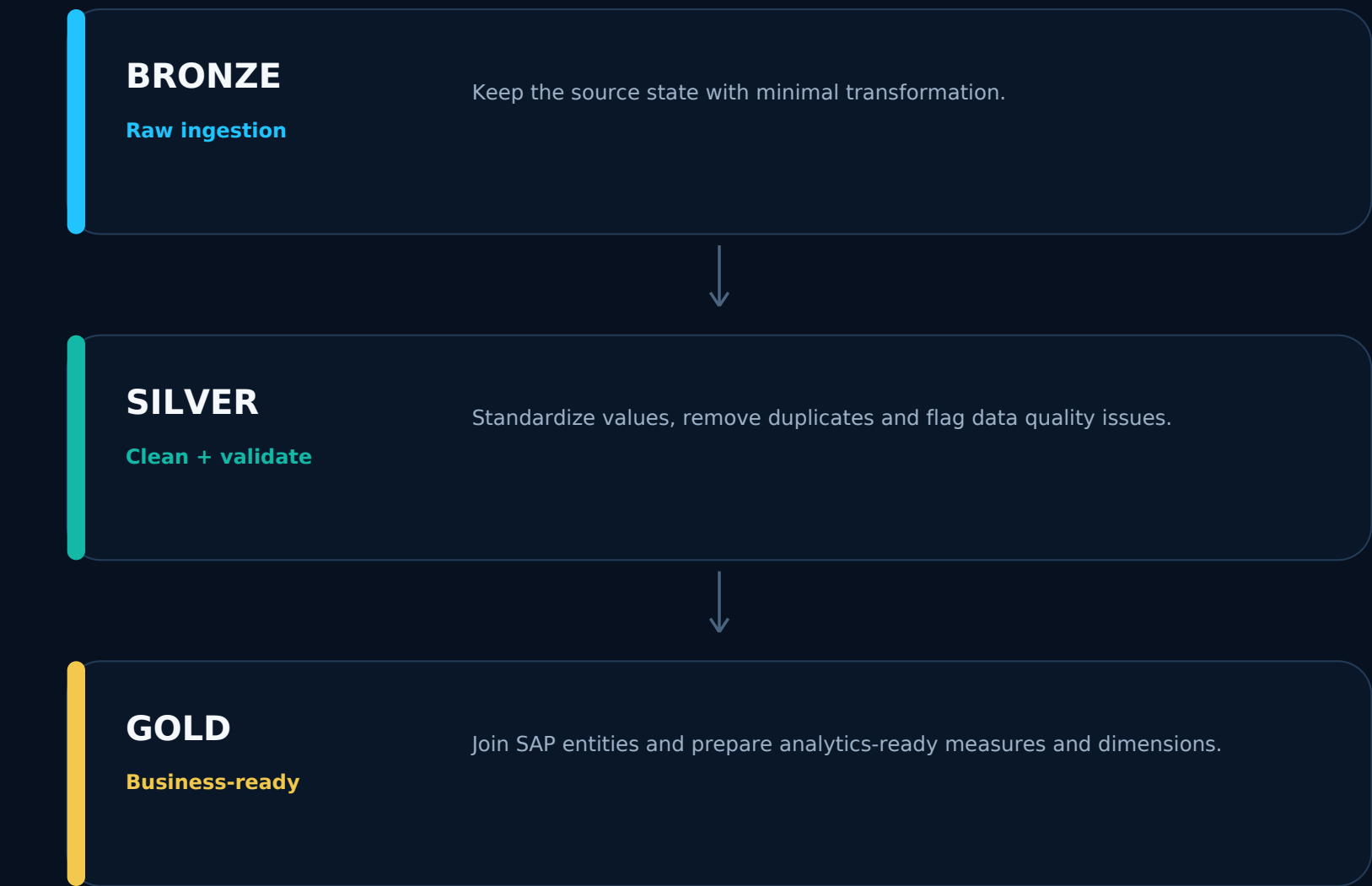
Project Architecture

A practical SAP SD use case built with Medallion Architecture

Source data



Medallion flow



Goal

Build a reusable sales analytics foundation for customers, materials, geography and time - then automate the full refresh.

Bronze Layer

Raw SAP-style data ingested into Unity Catalog managed Delta tables

What I implemented

- Created a dedicated catalog: sap_sd_project.
- Created bronze, silver and gold schemas.
- Loaded KNA1, MARA, VBAK and VBAP as raw Bronze tables.
- Kept dirty source values unchanged for traceability.
- Used managed Delta tables inside Unity Catalog.

Bronze objects

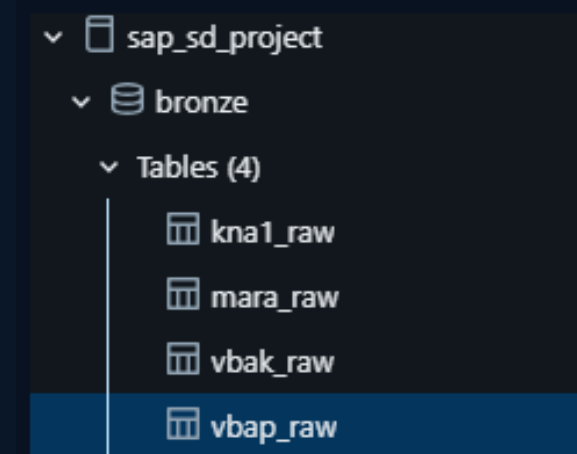
bronze.kna1_raw

bronze.mara_raw

bronze.vbak_raw

bronze.vbap_raw

Unity Catalog structure



Bronze stores the raw source. Cleaning and business validation are intentionally handled in Silver.

Why this matters

- Reproducibility - Silver and Gold can be rebuilt from the original source.
- Traceability - data quality issues can be compared against the untouched input.
- Separation of concerns - ingestion is separate from cleaning and business modeling.

Silver Layer - Customer & Material

PySpark cleaning and data quality engineering

KNA1 - Customer cleaning

- Trimmed customer names and standardized text formatting.
- Completed missing City / Region values using a controlled mapping.
- Standardized country values.
- Converted customer IDs to string and restored SAP-style leading zeros.
- Removed duplicates using MANDT + KUNNR as the business key.

Silver output

`silver.kna1_clean`

MARA - Material cleaning

- Trimmed MATNR and standardized material type values.
- Handled missing material groups with an explicit UNKNOWN category.
- Standardized units to EA.
- Flagged invalid negative gross weight and missing net weight.
- Validated net weight $\leq$ gross weight and removed duplicates.

`silver.mara_clean`

Silver Layer - Sales Orders

Header and item quality rules built with PySpark

VBAK - Sales Order Header

- Standardized currency, order status and sales organization.
- Parsed mixed date formats into a single clean ORDER_DATE.
- Flagged missing customers and invalid order dates.
- Standardized KUNNR and restored leading zeros.
- Removed duplicates using MANDT + VBELN.

`silver.vbak_clean`

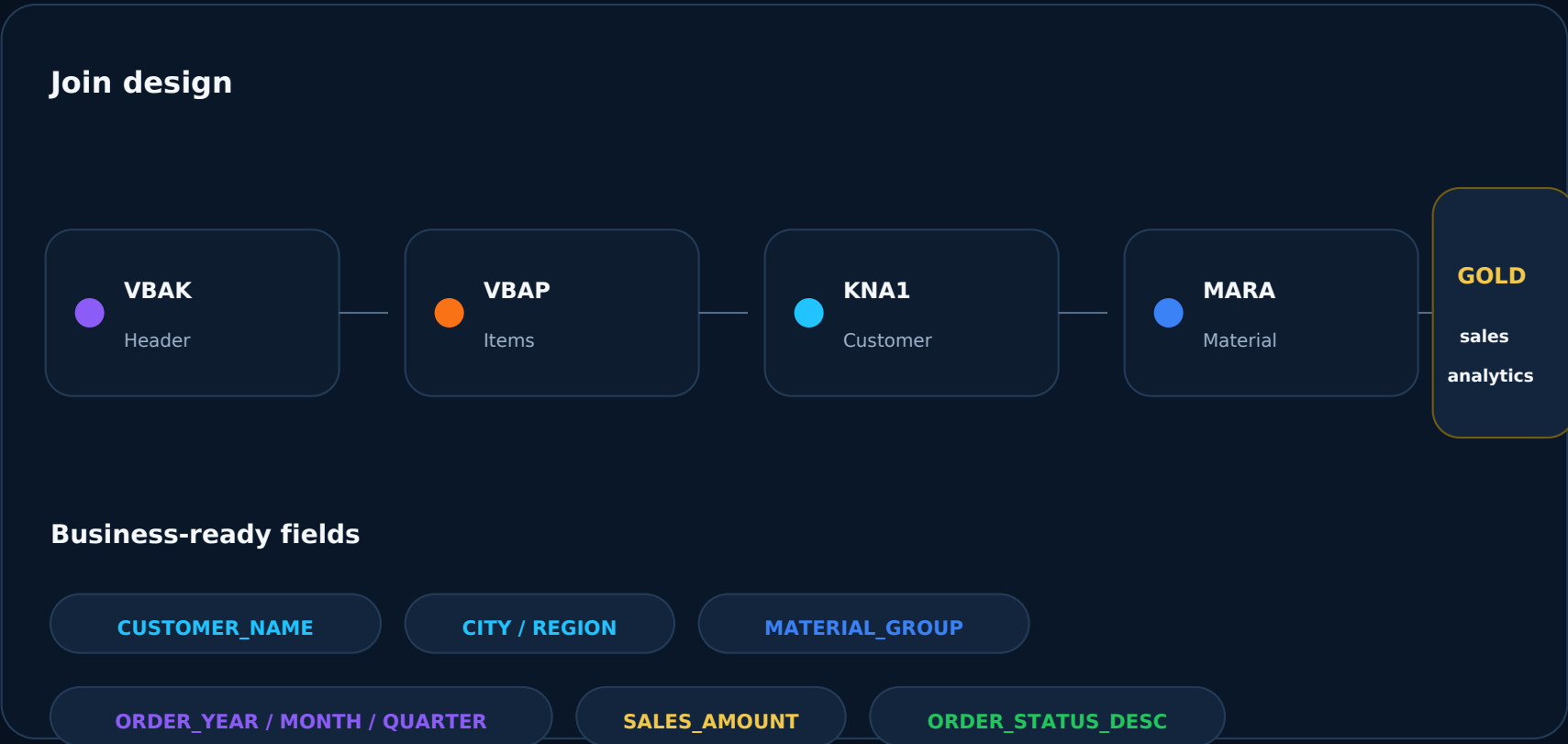
VBAP - Sales Order Items

- Standardized MATNR, sales units and plant values.
- Flagged invalid quantity and price values instead of silently correcting them.
- Validated NETWR against KWMENG x NETPR.
- Detected orphan material references with a left anti join.
- Removed duplicates using MANDT + VBELN + POSNR.

`silver.vbap_clean`

Gold Layer - Sales Analytics

A business-ready model built from the cleaned SAP SD entities



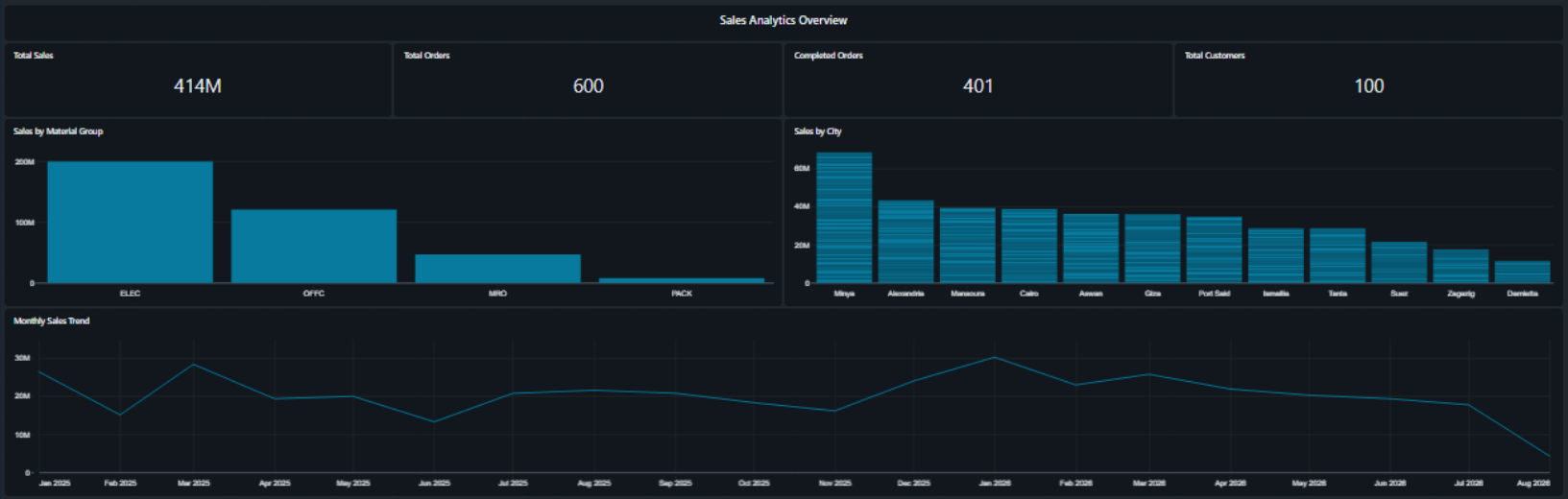
What the Gold layer adds

- Header + item data combined into one analytical sales grain.
- Customer master enrichment for customer, city, country and region analysis.
- Material master enrichment for material type and material group analysis.
- Time-derived fields: year, month and quarter.
- Readable order status descriptions and a reusable SALES_AMOUNT measure.

gold.sales_analytics

Analytics Dashboard

Final SAP SD sales dashboard built from the curated Gold layer



TOTAL SALES

414M

TOTAL ORDERS

600

COMPLETED ORDERS

401

TOTAL CUSTOMERS

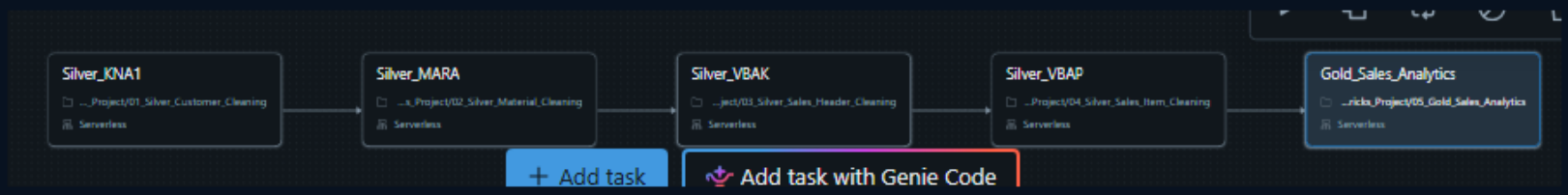
100

Dashboard focus

Material-group performance, city-level sales distribution and the monthly sales trend, supported by four executive KPIs.

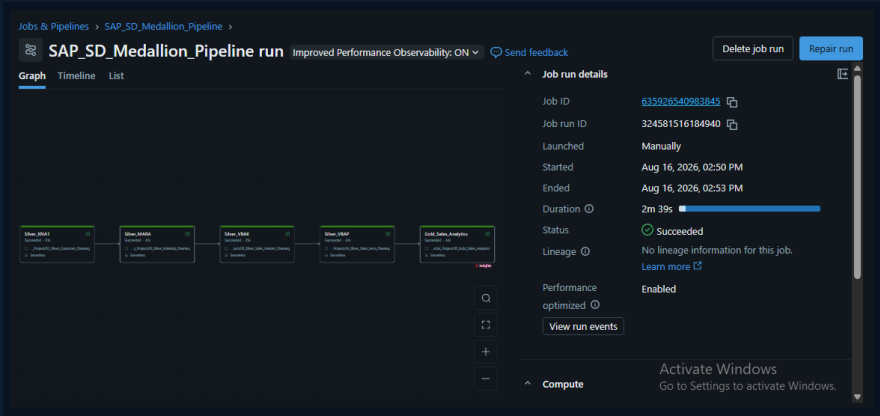
Orchestration, Scheduling & Monitoring

Lakeflow Jobs converted the notebooks into a repeatable end-to-end workflow



Dependency flow: Silver_KNA1 -> Silver_MARA -> Silver_VBAK -> Silver_VBAP -> Gold_Sales_Analytics

Successful run



STATUS

Succeeded

DURATION

2m 39s

Operational capabilities demonstrated

- Notebook task dependencies and ordered execution.
- Serverless execution.
- Daily scheduling.
- Run history and task-level monitoring.

Project Takeaways

What this project demonstrates as a Data Engineering portfolio piece

Capabilities demonstrated

- Medallion Architecture: Bronze -> Silver -> Gold
- Unity Catalog organization with catalog and schemas
- PySpark cleaning, standardization and validation
- Data quality flags, business-key deduplication and referential checks
- Delta tables for reusable managed data assets
- SAP SD entity joins and analytics-ready Gold modeling
- AI/BI dashboard creation from curated data
- Lakeflow Jobs orchestration, scheduling and monitoring

Portfolio summary

Built an end-to-end SAP SD data engineering solution on Databricks using PySpark, Delta, Unity Catalog, Medallion Architecture, AI/BI Dashboards and Lakeflow Jobs - from raw ingestion to a scheduled analytical refresh.

The project demonstrates both data engineering fundamentals and an SAP-oriented business context using customer, material and sales-order data.

SAP SD

PySpark

Delta

Unity Catalog

AI/BI

Lakeflow Jobs

Omar Oun Portfolio Project